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A physical tire model for real-time simulations

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ABSTRACT

We present a novel tire brush model with carcass flexibility, primarily designed for real-time applications while preserving the physical significance of its parameters. The tire's geometry is discretized and represented using a series of ribs, each intersecting the local road surface independently, on the other hand carcass in-plane fore-aft displacement, in-plane deflection, and out-of-plane sidewall torsion are approximated through a second-order polynomial. Forces and torques originating from the tire–road contact area are described using brush mechanics, where bristles' kinematics is directly linked to carcass deformation. Extensive effort is dedicated to understanding the possible sources of instability, as well as describing and implementing a robust numerical scheme to effectively solve the nonlinear system of equations arising from the modeling. Timing performance results are provided to demonstrate the suitability of the presented model for demanding hard real-time simulations. Finally, validation of the presented tire model is carried out by fitting experimental data and comparing it with the state-of-the-art MAGIC FORMULA model, proving its reliability and accuracy in reproducing tire behavior.

1. Introduction

Numerous investigations and experiments have been conducted over the past century to understand tire characteristics and behavior, revealing a close relationship between tire performance and the technologies, techniques, and materials used in manufacturing [17,45,46]. However, accurately modeling tire characteristics from construction specifications is challenging due to the complex tire internal structure. To tackle this challenge, two main modeling approaches have been proposed so far: *empirical* and *physical* [22,46,49,53].

The empirical approach is popular because of its simplicity, low computational cost, and ability to capture accurate tire behavior. This approach approximates tire–road interaction forces using a combination of functions derived from physical insights, as well as years of experimental campaigns and experience [46,49]. The identification process for computing best-fitting parameters given an experimental dataset is one of the biggest challenges for these models. The MAGIC FORMULA model, initially developed by Bakker et al. [2] and later improved multiple times [49], is the state-of-the-art empirical model [22,45], and it fits experimental data through a sequence of numerical optimizations on predetermined curve shapes [3]. Due to its diffusion, low computational complexity and overall performance, the MAGIC FORMULA model is considered the standard in both data fitting and vehicle simulation [22,49]. It is important to highlight that in its original versions, the MAGIC FORMULA does model neither thermodynamics nor wear effects. These phenomena significantly influence both tire and vehicle dynamics, particularly in high-performance scenarios [11,64]. Recognizing this, recent efforts have extended the MAGIC FORMULA model to encompass these effects, resulting in what is now known as the MF-evo model [12,76].

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The physical approach involves direct geometrical and physical tire modeling, providing a deeper understanding of the complex phenomena related to tire behavior [45]. While the computational burden heavily depends on the desired accuracy, this approach can better generalize behavior without fitting a large experimental dataset for every specific tire. Additionally, physical models usually depend on a smaller number of parameters that have a direct physical meaning. The Finite Element (FE) method represents a physical approach to tire modeling primarily used, though not exclusively, for assessing static characteristics, stiffnesses, resonant frequencies, and vibration modes [75]. A similar physical modeling technique is based on the Discrete Element (DE) method [33], which employs a limited number of interconnected elements and nodes to mimic the degrees of freedom and constraints of real tire carcass structures [15,18,81]. Prominent DE models like FTIRE[®] and CDTIRE[®] are capable of running under real-time environments [16,70], and encompass advanced features such as detailed external geometry description, flexible and visco-plastic rim, air cavity vibration, deformable and visco-elastic ground soil compatibility, as well as temperature and wear modeling. However, the availability of detailed information regarding the computational time performance of both FTIRE[®] and CDTIRE[®] remains limited.

A big branch of physical modeling relies on the brush mechanics theory [49], which originates from a simplified version of Kalker's conformal contact theory [28–32,41]. This theory assumes the road surface to be non-deformable, and describes the tire behavior within the contact patch area using one or more rows of radial bristles along or parallel to its equator, having a frictional constraint with the road, and deflecting according to kinematic equations. Over the years, several extensions of the original brush model have emerged, addressing dynamic friction behavior [8,9,35,78], realistic contact pressure distribution [13,43,80], high camber angles and high steering speeds [24,58], carcass deformation influence [17,26,27,42–44,56,57,74,80], effects of temperature and wear [13,23], and a more accurate description of external tire morphology [6,52]. Brush models are effective at efficiently capturing the behavior of the tire tread layer, and they can also be coupled to a simplified carcass deformation model under proper assumptions. Sakai's [59–62], NEO-FIALA [26,27,42–44], TREADSIM [7], and TAMETIRE[®] [13] are three notable advanced tire models that combine a brush model with a carcass deflection and, although different in design, they can accurately describe significant tire characteristics that are beyond the capabilities of empirical models. These include:

- advanced frictional properties at the tire–road interface, such as local contact pressure and velocity dependency of the friction coefficient;
- description of the tangential stress within the contact patch, crucial for understanding the tire's lateral and longitudinal force generation;
- carcass in-plane fore-aft displacement and deflection, and out-of-plane sidewall torsion, often described through a beam on elastic foundation or a second-order polynomial approximation.

Despite the extensive literature on tire modeling, the authors aim to develop a novel tire model suitable for real-time vehicle simulations, highlighting the scope for further enhancement in the field of tire modeling. This new model is built upon a minimal set of physical parameters rather than empirical formulas, to characterize tire properties. For this reason, the contributions of this paper to the existing state-of-the-art tire modeling include the development of a fully scalable physical tire model capable of running within a time step of 1 ms. Such a time step is a typical requirement for real-time simulations that aim at reproducing stiff subsystems with high-frequency dynamics [49]. The tire model is structured modularly, facilitating straightforward extension to incorporate additional features such as the effect of temperature and wear, and not less importantly, to achieve the desired computational performance through the newly developed numerical solution scheme. Possible sources of instability are also analyzed, and practical solutions to ensure robustness are discussed. In summary, this study aims to advance the current understanding of flexible carcass tire model behavior and provide a numerical framework enabling the newly developed physical tire model's integration into hard real-time simulations. To demonstrate the suitability of the proposed model in such demanding conditions, timing and numerical scheme performance outcomes are provided. Validation of the model is performed by fitting experimental data provided by the Tire Research Facility (TRF) for the Formula SAE Tire Test Consortium (FSAE TTC) testing program (refer to Kasprzak and Gentz [34] for a comprehensive description FSAE TTC's testing set-up, procedures, and data acquisition), as well as by comparing it with the state-of-the-art MAGIC FORMULA model, ensuring its reliability and accuracy in reproducing tire behavior.

The first section of the paper deals with tire–road contact and tire modeling (Section 2), which discusses tire and road geometric representation (Section 2.1), as well as tire–road vertical and tangential contact mechanics (Sections 2.2 and 2.3). The second part of the paper is dedicated to the development of an appropriate numerical solution approach for the tire model (Section 3). The third and last part of the paper delves into the proposed tire model's performance (Section 4) and presents timing results (Section 4.1) and discussions on the validation procedure (Section 4.2). The paper concludes with a summary of the main findings and future research directions (Section 5). Throughout the subsequent model description, the “ $\stackrel{\text{def}}{=}$ ” operator will be used. Its purpose is to avoid explicitly specifying the dependencies among the various quantities and to write the relationships in a more concise and readable manner.

2. Tire modeling

Pneumatic radial tires are inflated composite structures whose primary structural subcomponents are the *rim*, the *carcass*, and the *tread*. While the rim maintains tire shape, the carcass constitutes the main tire structure connecting the rim to the tread. The tread, on the other hand, serves as the outer rubber layer providing a link between the road and the carcass. In the following sections, we will describe the tire modeling approach, focusing on the carcass and tread, as the rim is considered a rigid body.

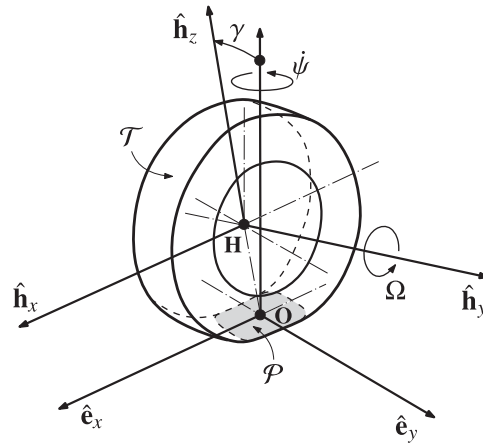


Fig. 1. Tire-road schematics according to ISO coordinate system [25].

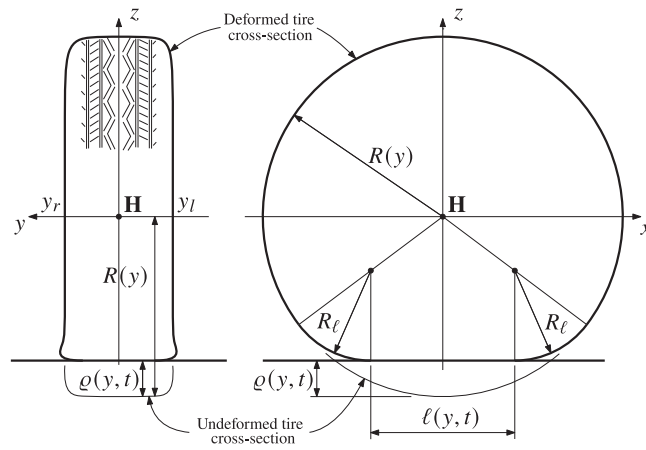


Fig. 2. Illustration of the tire-road contact geometry parameters.

2.1. Tire and road geometric representation

Let us consider a reference frame $\mathbf{H}xyz$ with unit vectors $(\hat{\mathbf{h}}_x, \hat{\mathbf{h}}_y, \hat{\mathbf{h}}_z)$, having its origin $\mathbf{H}$ located at the wheel hub. The axes are oriented according to the ISO 8855:2011 standard [25]. The x -axis is directed towards the longitudinal direction of motion, the z -axis points upward, and the y -axis is oriented so that the coordinate system is right-handed (refer to Fig. 1). Geometrically, the tire is characterized as axially symmetric (about the axis $\hat{\mathbf{h}}_y$), convex, and closed set $\mathcal{T}$ defined as

$$\mathcal{T} = \left\{ \begin{bmatrix} x \\ y \\ z \end{bmatrix} \in \mathbb{R}^3, y \in [y_l, y_r], R(y) : \mathbb{R} \mapsto \mathbb{R} \mid \sqrt{x^2 + z^2} \leq R(y) \right\}. \quad (1)$$

Here, the function $R(y)$ can be chosen arbitrarily to reconstruct the desired outermost shape of the tire. On the other hand, the road is modeled as a closed half-space $\mathcal{R}$ whose boundary is defined by the plane passing through the point $\mathbf{p}$ with unit normal $\hat{\mathbf{n}}$, i.e.,

$$\mathcal{R} = \{ \mathbf{x} \in \mathbb{R}^3 \mid (\mathbf{x} - \mathbf{p}) \cdot \hat{\mathbf{n}} \leq 0 \}, \quad (2)$$

where $\mathbf{x} = [x, y, z]^T$. The contact patch region is defined as a closed set $\mathcal{P}$ comprising all the points of the tire $\mathcal{T}$ which are also interior points of the road boundary surface $\partial\mathcal{R}$, i.e., $\mathcal{P} = \mathcal{T} \cap \partial\mathcal{R}$. The contact patch reference frame $\mathbf{O}xyz$ has the origin in $\mathbf{O}$, which corresponds to the centroid of the region $\mathcal{P}$. Its unit vectors $(\hat{\mathbf{e}}_x, \hat{\mathbf{e}}_y, \hat{\mathbf{e}}_z)$ are defined as $\hat{\mathbf{e}}_x = \hat{\mathbf{h}}_x$, $\hat{\mathbf{e}}_y = \hat{\mathbf{n}} \times \hat{\mathbf{e}}_x$, and $\hat{\mathbf{e}}_z = \hat{\mathbf{n}}$.

2.2. Tire-road vertical contact mechanics

When the contact patch set is not empty (i.e., $\mathcal{P} \neq \emptyset$) there is a multitude of contact points $\mathbf{x} = [x, y, z]^T \in \mathcal{P}$. At each point, the stress component along the z -axis (also known as normal stress) $q_z(\mathbf{x}, t)$ is applied. The integral of the normal stress distribution over

the entire contact patch region is equal to the tire's vertical load $F_z(t)$. The vertical normal stress distribution is extremely difficult to calculate analytically, and it depends on the tread pattern properties, carcass stiffness, friction and inflation pressure [45]. To easily model the tire's vertical characteristics, we employ a single-contact point approach. The total vertical tire stiffness is calculated as

$$k_z(\omega, \gamma, p) = k_z^s + k_z^p p + k_z^\omega \omega + k_z^\gamma \gamma^2. \quad (3)$$

Here, k_z^s represents the structural vertical stiffness, k_z^p denotes the vertical stiffness dependence on inflation pressure, k_z^ω indicates the vertical stiffness dependence on rolling speed, and k_z^γ is the vertical stiffness dependence on camber angle. The total vertical force is thereby calculated as

$$F_z \stackrel{\text{def}}{=} F_z(t) = k_z(\omega(t), \gamma(t), p)\rho(t) + c_z \dot{\rho}(t) + k_z^b \max(0, \rho(t) - \rho_b), \quad (4)$$

where $F_z \in \mathbb{R}_{\geq 0}$, $\rho(t) \in \mathbb{R}_{\geq 0}$ represents the deflection at the tire contact point, ρ_b represents the bottoming threshold, c_z is the vertical damping coefficient, and k_z^b indicates the bottoming vertical stiffness. Specifically, the tire contact point deflection $\rho(t)$ is calculated with the method outlined in [72,73]. This technique not only facilitates efficient and robust computation of the contact point deflection $\rho(t)$ but also allows us to compute the tire deflection $\rho \stackrel{\text{def}}{=} \rho(y, t)$ at a specific lateral coordinate y and time t . Alternatively, for cases where the technique outlined in [73] is not employed, a viable approximation, particularly for small camber angles ($|\gamma(t)| < \pi/10$), is

$$\rho(t) = \max_{y \in [y_l, y_r]} \rho(y, t) \quad (\text{refer to Fig. 2}). \quad (5)$$

2.2.1. Contact patch length

The geometrical intersection between the tire set $\mathcal{T}$ and road boundary surface $\partial\mathcal{R}$ provides us only a rough approximation of the tire–road contact patch area. Based on the findings in [36,51], we introduce a contact transition radius $R_\ell \in \mathbb{R}_{\geq 0}$ to take the tire belt stiffness into account and get a more realistic result (see Fig. 2). The contact length at any given lateral coordinate y on the contact patch reference system can be computed as

$$\ell \stackrel{\text{def}}{=} \ell(y, t) = 2\sqrt{(2(R(y) - R_\ell) - \rho(y, t))\rho(y, t)}, \quad (6)$$

where $\rho(y, t)$ is the tire radial deflection. Notice that Eq. (6) can be easily found from the study of Rhyne [51] by imposing null radial counter-deflection.

2.2.2. Contact pressure distribution

The pressure distribution is one of the most complex, challenging-to-model factors significantly impacting tire behavior [67]. Both experimental results and FE analyses reveal that the pressure trend over the contact patch is strongly three-dimensional and is influenced by numerous factors, including carcass structural stiffness, contact pressure, relative camber angle, tread pattern topology, and local road macro-roughness. Experimental evidence suggests that adequately inflated modern radial tires typically exhibit a plateau in the pressure distribution at the central part of the contact patch [45, Chapter 5]. Conversely, overinflated and underinflated radial tires may have more complicated distribution shapes [45,63]. Recent efforts have been aimed at more detailed modeling of the contact patch to better match experimental results [13,43,80]. An interesting approach involves modeling the contact pressure distribution using a quartic polynomial function $Y(\chi) = c_0 + c_1\chi + c_2\chi^2 + c_3\chi^3 + c_4\chi^4$. This function represents the normalized contact pressure distribution shape along the normalized longitudinal coordinate $\chi \in [0, 1]$. To determinate the polynomial coefficients $c_0, \dots, c_4$ the following constraints are imposed

$$\begin{cases} \frac{d^2}{d\chi^2} Y\left(\frac{1}{2}\right) = -\lambda & \text{convexity at the center of the contact patch } \chi = \frac{1}{2}, \\ \int_0^1 Y(\chi) d\chi = 1 & \text{normalization condition for the total contact force over the contact patch,} \\ \int_0^1 Y(\chi)\chi d\chi = \frac{1}{2} - \delta & \text{barycentre shift with respect to the center of the contact patch } \chi = \frac{1}{2}, \\ Y(0) = Y(1) = 0 & \text{absence of pressure at the edges of the contact patch } \chi = 0 \text{ and } \chi = 1. \end{cases} \quad (7)$$

Then, we solve for $c_0, \dots, c_4$, and the solution of the system is

$$c_0 = 0, \quad c_1 = -\frac{1}{3}\lambda + 10 + 60\delta, \quad c_2 = 2\lambda - 30 - 180\delta, \quad c_3 = -\frac{10}{3}\lambda + 40 + 120\delta, \quad \text{and} \quad c_4 = \frac{5}{3}\lambda - 20.$$

Here, $\lambda \in \mathbb{R}$ and $\delta \in \mathbb{R}$ respectively denote the convexity and longitudinal barycentre shift of the base curve $Y(\chi)$. Fig. 3 illustrates the behavior of the normalized pressure distribution function $Y(\chi)$ shape at different values of λ and δ . The only drawback of this approach is that for certain values of δ and λ , the condition of positive pressure $Y(\chi) \geq 0$ within the range $\chi \in [0, 1]$ is not always guaranteed and must be numerically checked in the software implementation. Thus, a careful selection of these parameters is essential. On the other hand, the use of a polynomial to describe the pressure distribution over the contact patch allows us to adopt some numerically robust algorithms and efficient evaluations for some of the calculations that will follow. Given the objective

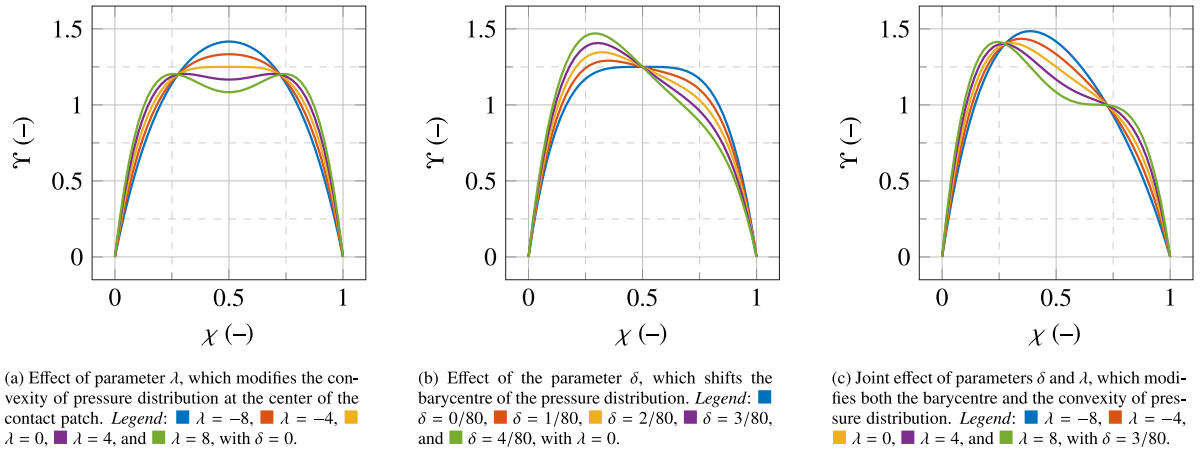


Fig. 3. Normalized pressure distribution function $Y(\chi)$ shape at different values of λ and δ .

of achieving real-time applicability for the numerical tire model, this aspect should not be overlooked. The vertical contact stress distribution $q_z(\mathbf{x}, t)$ is then written as

$$q_z \stackrel{\text{def}}{=} q_z(\mathbf{x}, t) = \frac{F_z(t)}{A_p(t)} Y\left(\frac{\ell(y, t)/2 - x}{\ell(y, t)}\right), \quad (8)$$

where $\mathbf{x} = [x, y, z]^T \in \mathcal{P}$, F_z denotes the tire's vertical load, and A_p represents the contact patch area, defined as

$$A_p \stackrel{\text{def}}{=} A_p(t) = \int_{y_r}^{y_l} \ell(y, t) dy. \quad (9)$$

It must be stressed that the contact patch region $\mathcal{P}$ is a function of time t and of the lateral coordinate y . This variability arises because the contact patch is not fixed but rather a function of the tire–road contact geometry.

2.3. Tire–road tangential contact mechanics

In the present paper, we assume that the vertical problem can be decoupled from the tangential, implying that the tangential stresses do not affect the vertical one, but not vice versa. Additionally, the vertical and lateral carcass flexibility characteristics are assumed to be fully independent. In simpler terms, we assume that neither friction-related phenomena nor carcass deformation affect the vertical pressure distribution q_z . Consequently, we can analyze tire–road tangential contact mechanics sequentially from the vertical component, while still maintaining the tangential contact forces proportionality to the vertical load F_z . It is worth noting that this assumption greatly simplifies the numerical solution of the tire model, as it avoids the need to find the vertical and tangential forces through a system of nonlinear equations, which would be computationally expensive.

2.3.1. Carcass deformation model

In numerous tire models, the carcass is modeled as a beam on elastic foundation [7,17,45,65]. Experimental findings from the studies [17,65] demonstrate that carcass deflection comprises both bending and shears components. The significance of the shear contribution varies depending on the carcass ply cord angles. Consequently, formulations like the Timoshenko–Ehrenfest or Euler–Bernoulli beam are particularly suited to describe the lateral bending of carcass structure in the cornering conditions [17]. Models like those in [13,14,17,26,27,43,59–62,80] adopt a second-order polynomial (parabola) as a local approximation of the beam on elastic foundation in the proximity of contact patch area. This approach provides a straightforward yet accurate description of the deformed carcass centerline within the contact region. However, the models currently present in the literature are only able to fully describe the lateral deformation and do not allow for modeling the longitudinal displacement. Indeed, during traction, braking and cornering, the contact patch is subject to non-negligible fore-aft, lateral, bending and torsional displacements. These displacements strongly impact the generation of longitudinal, lateral, and vertical forces, as well as rolling resistance, overturning, and self-alignment torques. To take into account the longitudinal displacement of the carcass, we define the two-dimensional carcass centerline as

$$\mathcal{L} \stackrel{\text{def}}{=} \mathcal{L}(\mathbf{x}, \mathbf{c}) = \begin{bmatrix} \mathcal{L}_x \\ \mathcal{L}_y \end{bmatrix} = \begin{bmatrix} x_c \\ y_c + \theta_c(x - x_c) - y_c \frac{\Psi}{2}(x - x_c)^2 \end{bmatrix}, \quad (10)$$

where time-dependent parameters x_c and y_c represent the carcass longitudinal and the lateral translating deformations, respectively, θ_c denotes the carcass twisting angle, and Ψ is the lateral bending shape factor. Fig. 4 illustrates the carcass baseline behavior at different values of parameters x_c , θ_c , and Ψ . As depicted in Fig. 5, we model the carcass deformation through a point $\mathbf{c} = [x_c, y_c, \theta_c]^T$

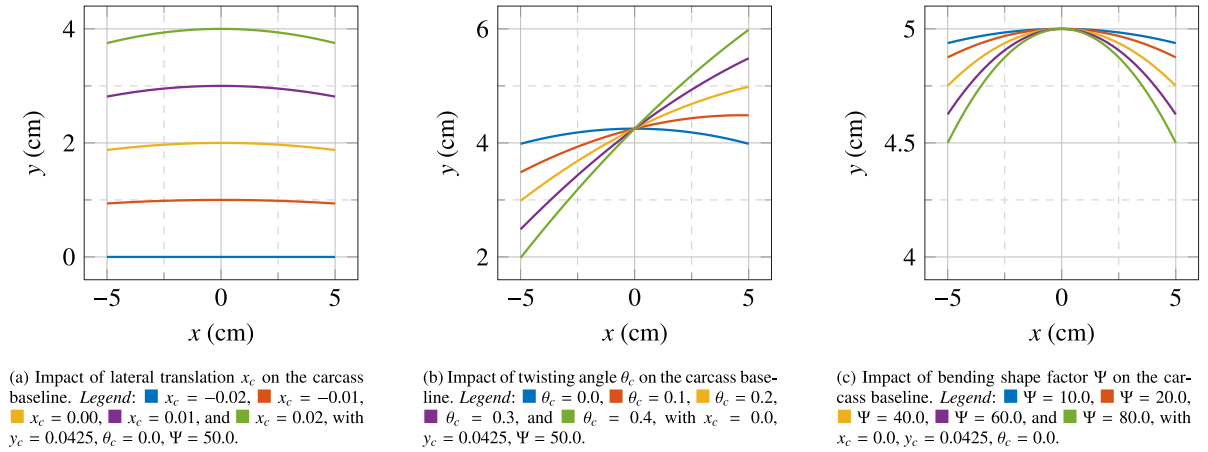


Fig. 4. Carcass baseline behavior at different values of parameters y_c , θ_c , and Ψ , with $\ell = 0.1$. The parameter x_c , as it only induces a shift along the x -axis of the baseline curve $\mathcal{L}(\mathbf{x}, \mathbf{c})$.

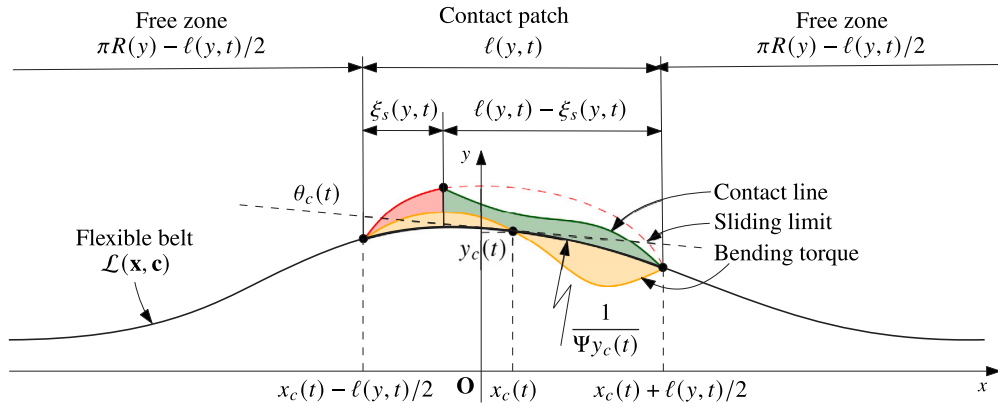


Fig. 5. Depictions of the deformed carcass baseline $\mathcal{L}(\mathbf{x}, \mathbf{c})$, along with the contact line of bristles, adherence limit, distribution of bending torque, and the adherence and sliding regions.

described within the contact patch reference frame $\mathbf{O}_{xyz}$. The carcass point $\mathbf{c}$ is connected through a set of independent spring elements to the point $[0, 0, -R_z]^\top$, defined in the hub reference frame $\mathbf{H}_{xyz}$, where $R_z = \hat{\mathbf{n}} \cdot (\mathbf{H} - \mathbf{O})$.

As mentioned at the beginning of this, the carcass is the subcomponent of a tire that connects the rim to the tread and, it has a significant impact on the tire's overall behavior. From a pure physics perspective, the carcass is a system that is rigidly connected to the rim and has to deform in order to accommodate the tread's deformation. This involves balancing forces generated by the carcass state against the stresses of the tread state deformation. Thence, the carcass equilibrium equations in matrix form are written as

$$\mathbf{G}^{\text{def}}(\mathbf{c}) = \mathbf{K}(\mathbf{c})\mathbf{c} - \mathbf{F}(\mathbf{c}, t) = 0, \quad (11)$$

where

$$\mathbf{F}^{\text{def}} = \mathbf{F}(\mathbf{c}, t) = \begin{bmatrix} F_{ax}(\mathbf{c}, t) \\ F_{ay}(\mathbf{c}, t) \\ M_{az}(\mathbf{c}, t) \end{bmatrix} + \begin{bmatrix} F_{sx}(\mathbf{c}, t) \\ F_{sy}(\mathbf{c}, t) \\ M_{sz}(\mathbf{c}, t) \end{bmatrix}, \quad (12)$$

and the matrix $\mathbf{K}$ represents the carcass stiffness matrix, which is assumed to be a diagonal matrix of the form

$$\mathbf{K}^{\text{def}} = \mathbf{K}(\mathbf{c}) = \begin{bmatrix} K_x(p, x_c) & 0 & 0 \\ 0 & K_y(p, y_c) & 0 \\ 0 & 0 & K_\theta(p, \theta_c) \end{bmatrix}. \quad (13)$$

The entries $K_x(p, x_c)$, $K_y(p, y_c)$, and $K_\theta(p, \theta_c)$ are the longitudinal, lateral and twisting carcass stiffnesses, respectively. These values are influenced by the tire construction technology and inflation pressure p , as well as the deformation of the tire carcass. They can

be expressed as follows

$$\begin{aligned} K_x(p, x_c) &= k_x^s + p k_x^p + k_x^b h(x_c, x_b, h_x), \\ K_y(p, y_c) &= k_y^s + p k_y^p + k_y^b h(y_c, y_b, h_y), \\ K_\theta(p, \theta_c) &= k_\theta^s + p k_\theta^p + k_\theta^b h(\theta_c, \theta_b, h_\theta). \end{aligned} \quad (14)$$

Here, the function $h(x, b, h)$ is defined as

$$h(x, b, h) = (\text{pos}(-x - b, h) + \text{pos}(x - b, h))^2, \quad \text{where} \quad \text{pos}(x, h) = \begin{cases} \frac{\sqrt{x^2 + h^2}}{2} + x & x > 0 \\ \frac{h^2}{2(\sqrt{x^2 + h^2} - x)} & \text{otherwise} \end{cases} \quad (15)$$

is a regularized function of the positive part. Notice that the deformation of the tire carcass has a physical limit given by the maximum deformation of the sidewalls. For this reason, the parameters k_x^b , k_y^b , and k_θ^b are introduced to model the longitudinal, lateral and twisting bottoming stiffnesses, respectively.

2.3.2. Tire–road tangential contact mechanics equations

For each point $\mathbf{x} \in \mathcal{P}$, given a carcass pose $\mathbf{c}$, we associate a speed field $\mathbf{v} \stackrel{\text{def}}{=} \mathbf{v}(\mathbf{x}, t)$ and a displacement vector $\mathbf{u} \stackrel{\text{def}}{=} \mathbf{u}(\mathbf{x}, \mathbf{c}, t)$. The speed field $\mathbf{v}$ represents the relative speed between the outer tread layer and the road, while the displacement vector $\mathbf{u}$ represents the deformation of the material point (also referred to as a *bristle*) located at the coordinate $\mathbf{x}$. Additionally, each material point is also subjected to an external force $\mathbf{q} \stackrel{\text{def}}{=} \mathbf{q}(\mathbf{x}, \mathbf{c}, t)$ generated by the tire–road contact.

We define the generic tangential position as $\mathbf{t} = x\hat{\mathbf{e}}_x + y\hat{\mathbf{e}}_y$, the tangential velocity field as $\mathbf{v}_t \stackrel{\text{def}}{=} \mathbf{v}_t(\mathbf{x}, \mathbf{c}, t) = v_x(\mathbf{x}, \mathbf{c}, t)\hat{\mathbf{e}}_x + v_y(\mathbf{x}, \mathbf{c}, t)\hat{\mathbf{e}}_y$, the tangential displacement vectors as $\mathbf{u}_t \stackrel{\text{def}}{=} \mathbf{u}_t(\mathbf{x}, \mathbf{c}, t) = u_x(\mathbf{x}, \mathbf{c}, t)\hat{\mathbf{e}}_x + u_y(\mathbf{x}, \mathbf{c}, t)\hat{\mathbf{e}}_y$, and the tangential stress vector as $\mathbf{q}_t \stackrel{\text{def}}{=} \mathbf{q}_t(\mathbf{x}, \mathbf{c}, t) = q_x(\mathbf{x}, \mathbf{c}, t)\hat{\mathbf{e}}_x + q_y(\mathbf{x}, \mathbf{c}, t)\hat{\mathbf{e}}_y$, in a such a way that

$$\mathbf{v} = \begin{bmatrix} \mathbf{v}_t \\ v_z \end{bmatrix}, \quad \mathbf{u} = \begin{bmatrix} \mathbf{u}_t \\ u_z \end{bmatrix}, \quad \text{and} \quad \mathbf{q} = \begin{bmatrix} \mathbf{q}_t \\ q_z \end{bmatrix}. \quad (16)$$

The bristles tangential *micro-sliding speed* is defined as

$$\mathbf{v}_s \stackrel{\text{def}}{=} \mathbf{v}_s(\mathbf{x}, \mathbf{c}, t) = \begin{bmatrix} v_{sx}(\mathbf{x}, \mathbf{c}, t) \\ v_{sy}(\mathbf{x}, \mathbf{c}, t) \end{bmatrix}, \quad (17)$$

which represents the relative speed between a material point within the contact patch $\mathcal{P}$ and the road surface $\partial\mathcal{R}$. In the following, we will employ an isotropic steady-state Coulomb friction model, which enables us to write the equations governing the contact mechanics between the tire tread layer and the road surface as

$$\mathbf{v}_s = 0 \iff \|\mathbf{q}_t\| \leq \mu_s q_z \quad \text{and} \quad \mathbf{q}_t = -\mu_d q_z \frac{\mathbf{v}_s}{\|\mathbf{v}_s\|} \iff \mathbf{v}_s \neq 0. \quad (18)$$

In these equations, μ_s represents the static friction coefficient and $\mu_d \stackrel{\text{def}}{=} \mu_d(\mathbf{x}, \mathbf{c}, t)$ denotes the “dynamic” friction coefficient. It is important to note that (18) is only valid under the assumption of memory-less friction, meaning there is a one-to-one map between the tangential stress and the micro-sliding speed [79]. To solve (18), two additional sets of equations are necessary. The first set consists of the so-called *tire–road kinematic equations*, which link the wheel hub’s relative motion to the deformation field over the contact patch area. The second set includes the *constitutive relations*, which correlate the bristles deformation field to the generated tangential stress $\mathbf{q}_t$.

2.3.3. Rubber friction

Friction significantly affects tire behavior. As noted in prior works [66,68,69,77], it is influenced by various factors such as tread rubber compound, road surface roughness at both micro and macro levels, relative speed of sliding surfaces, and temperature conditions. To keep the complexity under an acceptable level, we adopt a steady-state Coulomb friction model. In doing so, the dynamic characteristic of friction, known as *frictional memory*, is disregarded. Such a behavior arises from the fact that the friction coefficient depends not only on the relative sliding speed but also on the contact history, i.e., the sliding trajectory [50]. Including this phenomenon would significantly increase the complexity of the whole tire model, necessitating a dense discretization of the tire tread layer.

Numerous physical formulations aim to capture the relationship between the friction coefficient and sliding speed. One of the most important is the *Savkoor friction law*, which includes the so-called *Stribeck effect* [66–68]. When accounting for contact pressure dependency, the “dynamic” friction coefficient $\mu_d(\mathbf{x})$ can be expressed as

$$\mu_d = \mu_w(\mu_{p_0} + \mu_{p_1} \exp(-\mu_{p_2} q_z))(\mu_k + (\mu_s - \mu_k) \exp\left(-\mu_{v_0}^2 \log^2\left(\frac{\|\mathbf{v}_s\|}{v_\mu} + \mu_{v_1} \exp\left(-\frac{\|\mathbf{v}_s\|}{v_\mu}\right)\right)\right), \quad (19)$$

where

- μ_w represents the friction coefficient scaling factor, used to tune the friction coefficient under different conditions, such as wet, dry, icy, etc.;

- μ_s is the static friction coefficient;
- μ_k is the kinetic friction coefficient;
- $\mu_{p0}, \mu_{p1}, \mu_{p2}$ are parameters aimed at fitting the rubber-asphalt contact friction pressure dependency;
- μ_{v0} and μ_{v1} are parameters used to reproduce the rubber-asphalt contact friction velocity dependency;
- v_μ is the Stribeck velocity.

2.3.4. Tire–road kinematic equations

The relative velocity between the bristles and the road, first introduced with the name of micro-sliding velocity $\mathbf{v}_s$, are expressed in Eulerian form similar to that in [22,37,49,55–58]. Hence, using (10) and (16), $\mathbf{v}_s$ can be written as

$$\mathbf{v}_s = V_r \left(\boldsymbol{\sigma} + \varphi \hat{\mathbf{e}}_z \times (\mathcal{L} + \mathbf{t}) + \frac{d}{dx} \mathcal{L} \right) + \frac{d}{dt} \mathbf{u}_t, \quad (20)$$

where the material point deformation $\mathbf{u}_t$ is assumed to be small. For a comprehensive description and derivation of such sliding refer to [22,37,49,53,55,57,58]. The *theoretical translational slips* $\boldsymbol{\sigma}$ represent the normalized difference between the longitudinal and lateral components of the speed of the wheel hub and that of the tire periphery $V_r \stackrel{\text{def}}{=} V_r(y, t) = \omega(t)R_r(y, t)$. Therefore, they are defined as

$$\boldsymbol{\sigma} = \begin{bmatrix} \sigma_x \\ \sigma_y \end{bmatrix} \stackrel{\text{def}}{=} \boldsymbol{\sigma}(y, t) = \begin{bmatrix} \sigma_x(y, t) \\ \sigma_y(y, t) \end{bmatrix} = -\frac{1}{|V_r(y, t)|} \begin{bmatrix} V_x(t) - V_r(y, t) + \dot{x}_c(t) \\ V_y(t) - R_z(t)\dot{\gamma}(t) + \dot{y}_c(t) \end{bmatrix}, \quad (21)$$

where σ_x and σ_y are referred to as the *longitudinal* and the *lateral theoretical slips*. The *theoretical spin slip* φ is obtained from

$$\varphi \stackrel{\text{def}}{=} \varphi(y, t) = -\frac{\dot{\psi}(t) + \omega(t) \sin(\gamma(t)) + \dot{\theta}_c(t)}{|V_r(y, t)|}. \quad (22)$$

The quantity R_r represents the so-called *real rolling radius*, which is calculated as

$$R_r \stackrel{\text{def}}{=} R_r(y, t) = R(y) - \frac{\varphi(y, t)}{3}. \quad (23)$$

Other definitions of slips are available in the literature, such as the *practical longitudinal slip* κ and *slip angle* α [49], which are defined as

$$\begin{bmatrix} \kappa \\ \alpha \end{bmatrix} \stackrel{\text{def}}{=} \begin{bmatrix} \kappa(y, t) \\ \alpha(y, t) \end{bmatrix} = -\frac{1}{|V_x(t)|} \begin{bmatrix} V_x(t) - V_r(y, t) + \dot{x}_c(t) \\ \arctan V_y(t) - R_z(t)\dot{\gamma}(t) + \dot{y}_c(t) \end{bmatrix}, \quad \text{and thus} \quad \begin{bmatrix} \sigma_x \\ \sigma_y \end{bmatrix} = \frac{1}{1 + |\kappa|} \begin{bmatrix} \kappa \\ -\tan(\alpha) \end{bmatrix}. \quad (24)$$

Since we are using the same Eulerian form of [58], the total time derivative is given by

$$\frac{d}{dt} = \frac{\partial}{\partial t} + \mathbf{v}_t \cdot \nabla_t, \quad (25)$$

where ∇_t collects the tangential components of the gradient, and $\partial/\partial t = 0$ since we are neglecting the viscous component of the Kelvin-Voigt element representing the rubber material point [40]. Neglecting the viscous effect of the tread rubber material eliminates any explicit time dependence in the formulation of brush model forces and torques. Consequently, the tread rubber always operates under stationary conditions. Considering these points, the total derivative of the material point can be explicitly expressed as

$$\frac{d}{dt} \mathbf{u}_t = \underbrace{\frac{\partial}{\partial t} \mathbf{u}_t}_{=0} + \mathbf{v}_t \cdot \nabla_t \mathbf{u}_t, \quad (26)$$

where the tangential gradient is given by $\nabla_t = [\partial/\partial x, \partial/\partial y]^\top$. The velocity field $\mathbf{v}_t$ of the tread rubber is formulated as

$$\mathbf{v}_t = -V_r \frac{\frac{d}{dx} \mathcal{L}}{\left\| \frac{d}{dx} \mathcal{L} \right\|}. \quad (27)$$

Given the assumption of small carcass deformations (approximately $|x_c| < \rho_b$, $|y_c| < \rho_b$, and $|\theta_c| < \pi/10$) and small tire camber angles ($|\gamma(t)| < \pi/10$) the following relationships holds [54]

$$\frac{d}{dx} \mathcal{L}_x \approx 1 \quad \text{and} \quad \frac{d}{dx} \mathcal{L}_y \approx 0. \quad (28)$$

The trajectories of the bristles moving within the contact patch are straightened in the longitudinal direction. Therefore, the formulation is constrained to a velocity field of the form $\mathbf{v}_t = [-V_r, 0]^\top$

By separating the longitudinal and tangential displacements of the bristles, we derive the bristle micro-sliding speed formulated as

$$\mathbf{v}_s = V_r \left(\boldsymbol{\sigma} + \varphi \hat{\mathbf{n}}_z \times (\mathcal{L} + \mathbf{t}) + \frac{d}{dx} \mathcal{L} - \frac{d}{dx} \mathbf{u}_t \right). \quad (29)$$

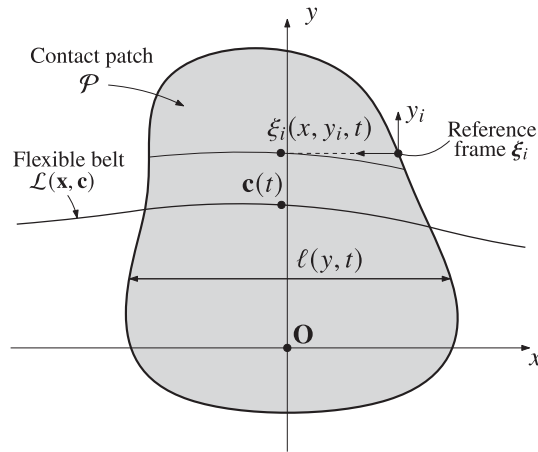


Fig. 6. Representation of the contact patch region $\mathcal{P}$ and contact patch reference frame $\xi_i(y_i, t)$. Notice that the contact patch is defined as a D-convex and arbitrary region, whose shape is described along the y -axis by the quantity $\ell(y, t)$.

2.3.5. Constitutive relations

In the present paper, we assume that – from a frictional perspective – two regions are identified along the contact patch region $\mathcal{P}$. The first is the adhesion region $\mathcal{A}$, where bristle tips do not slide with respect to the ground, generating the so-called adhesive forces. The second is the sliding region $\mathcal{S}$, where bristles tips slide on the ground surface, producing instead the sliding forces. The boundary separating these two regions is called transition line $\xi_s \stackrel{\text{def}}{=} \xi_s(\xi, \mathbf{c}, t)$. Although this assumption may be not realistic in the case of high spin slip value, it greatly simplifies the problem [54]. By adopting this approach, the domain corresponds to the contact patch interior $\tilde{\mathcal{P}}$, where the partial differential equations ruling the tire–road kinematics are defined, and the corresponding boundary conditions can be uniquely formulated on $\partial\mathcal{P}$ [58]. This research aims to develop a formulation that strikes a balance between realism and computational efficiency, rather than striving for a rigorous closed-form expression for the tangential bristle contact mechanics.

Adhesion region. Each individual, material point, acting independently of others, enters the contact patch through the leading edge in an undeformed state. As a result of van der Waals molecular bonding and rubber indentation at the rubber/ground interface, the tip of the bristles sticks to the ground. However, due to the micro-sliding velocity $\mathbf{v}_s$ between the root of the bristles and the road, a tangential deflection $\mathbf{u}_t$ starts to build up, along with tangential stress $\mathbf{q}_t$. Initially null, this bristle deformation increases gradually along the contact length until it reaches the transition line $\xi_s \stackrel{\text{def}}{=} \xi_s(\xi, \mathbf{c})$. This transition line represents the contact length coordinate at which the local force exerted by the bristle deformation in adhesion equals the maximum possible static friction force.

It is important to note that within the adhesion region, the tips of the bristles do not slide relative to the ground, resulting in a zero micro-sliding velocity. Consequently, the deformation of the bristles increases along the contact length direction with a spatial derivative equal to

$$\frac{d}{dx} \mathbf{u}_t = \frac{d}{dx} \begin{bmatrix} u_x \\ u_y \end{bmatrix}, \quad (30)$$

which is easily found from (29) by imposing $\mathbf{v}_s = 0$.

We now introduce a new contact patch coordinate system (also depicted in Fig. 6)

$$\xi = \begin{bmatrix} \xi \\ \eta \\ \zeta \end{bmatrix} \stackrel{\text{def}}{=} \xi(y, t) = \begin{bmatrix} \xi(x, y, t) \\ \eta \\ \zeta \end{bmatrix} = \begin{bmatrix} \ell(y, t)/2 - x \\ y \\ z \end{bmatrix}. \quad (31)$$

The variable ξ represents the distance from the entrance to the longitudinal coordinate, measured longitudinally from the contact patch leading edge. From now on, we will use the superscript “ ξ ” to denote variables described in reference system ξ , e.g., $\mathbf{u}_t^\xi \stackrel{\text{def}}{=} \mathbf{u}_t(\xi, \mathbf{c})$. Substituting ξ in (30) we obtain

$$\frac{d}{d\xi} \mathbf{u}_t^\xi = \frac{d}{d\xi} \begin{bmatrix} u_x^\xi \\ u_y^\xi \end{bmatrix}. \quad (32)$$

The specific deformation $\mathbf{u}_t^\xi$ at a contact length coordinate ξ and lateral coordinate y is obtained from integrating (32) over space, while imposing the boundary condition $\mathbf{u}_t(\mathbf{x} = [0, y, 0]^\top, \mathbf{c}) = 0$, namely

$$\mathbf{u}_t^\xi = \int_0^\xi \frac{d}{d\zeta} \mathbf{u}_t(\zeta, \mathbf{c}) d\zeta. \quad (33)$$

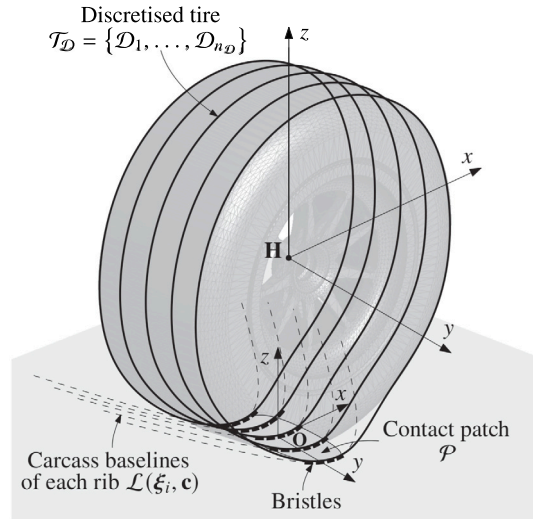


Fig. 7. Illustration of the brush model. The tire $\mathcal{T}$ is discretized into a series of n_D ribs. Carcass and bristles are represented as the main components of the model. Conversely, the rim is assumed to be a rigid body and thus has no impact on the tire–road contact.

The infinitesimal adherence force at a specific contact length coordinate ξ is determined by the product between (33) and the tread tangential stiffness matrix $\mathbf{k}$

$$\mathbf{q}_t^\xi = -\nu \mathbf{K}_t \mathbf{u}_t^\xi. \quad (34)$$

Here, ν represents the tread pattern void ratio, and $\mathbf{K}_t$ denotes the tangential stiffness matrix. This latter is typically expressed as

$$\mathbf{K}_t = \begin{bmatrix} k_{xx} & k_{yx} \\ k_{yx} & k_{yy} \end{bmatrix}, \quad (35)$$

where $k_{xy} \approx k_{yx} \ll k_{xx} \approx k_{yy}$, as proofed by Okonieski et al. [47]. The total bristles force induced by the deformation in adherence is obtained by integrating the following over the entire adhesion region $\mathcal{A}$

$$\mathbf{F}_a \stackrel{\text{def}}{=} \mathbf{F}_a(\mathbf{c}, t) = \int_{\mathcal{A}} \mathbf{q}_t^\xi d\xi. \quad (36)$$

In the original brush model presented in [49], the self-aligning moment is attributed solely to the asymmetrical distribution of lateral shear stress along the contact length. However, in this study, we will also consider that the difference in longitudinal force on each side of the contact patch contributes to the generation of the self-aligning moment. Thereby, the adhesive self-aligning moment is defined as the cross-product between forces induced by deformation and the position vector (the sum of bristle position and carcass baseline position vectors), which reads

$$\mathbf{M}_a \stackrel{\text{def}}{=} \mathbf{M}_a(\mathbf{c}, t) = \int_{\mathcal{A}} \mathbf{q}_t^\xi \times (\mathcal{L}^\xi + \xi) d\xi. \quad (37)$$

Slide region. When the local force induced by bristle deformation in adhesion is greater than the local maximum explicable static friction force, the bristle tips begin to slide over the ground with a micro-sliding velocity $\mathbf{v}_s^\xi \neq 0$. The local force exerted by bristle deformation in sliding opposes the micro-sliding velocity. Thus, its direction is denoted as

$$\mathbf{d}_t^\xi = \begin{bmatrix} d_{tx}^\xi \\ d_{ty}^\xi \end{bmatrix} \stackrel{\text{def}}{=} \mathbf{d}_t(\xi, \mathbf{c}) = \begin{bmatrix} d_{tx}(\xi, \mathbf{c}) \\ d_{ty}(\xi, \mathbf{c}) \end{bmatrix} = -\frac{\mathbf{v}_s^\xi}{\|\mathbf{v}_s^\xi\|}. \quad (38)$$

The total bristles force induced by the deformation in sliding is determined by integrating the following expression over the entire sliding region $\mathcal{S}$

$$\mathbf{F}_s \stackrel{\text{def}}{=} \mathbf{F}_s(\mathbf{c}, t) = \int_{\mathcal{S}} \mu_d^\xi q_z^\xi \mathbf{d}_t^\xi d\xi. \quad (39)$$

Similar to the adhesive scenario, the self-aligning moment is defined as the cross-product between the local force induced by the sliding and the position vector

$$\mathbf{M}_s \stackrel{\text{def}}{=} \mathbf{M}_s(\mathbf{c}, t) = \int_{\mathcal{S}} \mu_d^\xi q_z^\xi \mathbf{d}_t^\xi \times (\mathcal{L}^\xi + \xi) d\xi. \quad (40)$$

3. A numerical solution approach

Due to the complexity of the problem outlined in the preceding sections, finding an analytical solution is not feasible. Therefore, we need to further analyze the contact zone between the tire and the road and set some more assumptions that will allow us to obtain an effective numerical scheme.

3.1. Tire–road contact discretization

An essential characteristic of the contact patch set $\mathcal{P}$ is its constant compactness and D-convexity along the $\hat{\mathbf{e}}_x$ direction [39,58]. Leveraging the D-convexity properties enables us to discretize the contact patch into longitudinal sections. To optimize the intersection of the tire and road sets, we discretized the tire set $\mathcal{T}$ into a series of n_D ribs, similarly to the approach in [6,73]. This discretization is particularly useful for the numerical solution of the contact mechanics problem, as it allows us to scale the dimensionality of the problem to the number of ribs, which can be adjusted to balance the computational cost, as well as the accuracy of the solution.

Each tire rib, denoted as D , is characterized as a conical frustum which is located in $\mathbf{o} = [0, y_D, 0]^T$ in the $\mathbf{H}_{xyz}$ coordinate system, with a radius of $R(y_D)$ and a width of w_D

$$D = \left\{ w_D, y_D \in \mathbb{R}, \mathbf{x} = [x, y, z]^T \in \mathbb{R}^3 \mid \sqrt{x^2 + z^2} \leq R(y_D), |y - y_D| \leq \frac{w_D}{2} \right\}. \quad (41)$$

This new representation allows us to build the discretized tire set $\mathcal{T}_D$ as

$$\mathcal{T}_D = \left\{ D_1, D_2, \dots, D_{n_D-1}, D_{n_D} \right\}. \quad (42)$$

Notice that we will employ a set of contact patch coordinate systems ξ_i equal to the ribs n_D (see Figs. 6 and 7). These coordinate systems are redefined as

$$\xi_i = \begin{bmatrix} \xi_i \\ \eta_i \\ \zeta \end{bmatrix} \stackrel{\text{def}}{=} \begin{bmatrix} \xi_i(x, y_i, t) \\ \eta_i \\ \zeta \end{bmatrix} = \begin{bmatrix} \ell(y_i, t)/2 - x \\ y_i \\ z_i(x) \end{bmatrix}, \quad (43)$$

where $z_i(x)$ represents the height of the rib i at the point x , considering the possibility of the contact patch being sloped in the longitudinal direction. Similar to our previous approach, the variables described in the reference system ξ_i will be denoted with the subscript “ ξ_i ”, e.g., $\mathbf{u}_i^{\xi_i} \stackrel{\text{def}}{=} \mathbf{u}_i(\xi_i, c)$. It is worth noting that this tire representation is fully adjustable in scale, allowing for the density of ribs representing the tire to be tailored based on the required accuracy and computational speed for a specific simulation.

3.2. Discretized constitutive relations

By applying the considerations and assumptions outlined in Section 2.3.5 to the constitutive relations, we can express the integrals of forces and moments relative to the adhesion and sliding zones as summations of the components from each rib, specifically

- Adhesion forces

$$F_{xa}(\mathbf{c}, t) = \sum_{i=1}^{n_D} w_i \int_0^{\xi_i^s} q_x^{\xi_i} d\xi_i, \quad (44a)$$

$$F_{ya}(\mathbf{c}, t) = \sum_{i=1}^{n_D} w_i \int_0^{\xi_i^s} q_y^{\xi_i} d\xi_i, \quad (44b)$$

$$F_{za}(\mathbf{c}, t) = \sum_{i=1}^{n_D} w_i \int_0^{\xi_i^s} q_z^{\xi_i} d\xi_i, \quad (44c)$$

- Adhesion moments

$$M_{xa}(\mathbf{c}, t) = \sum_{i=1}^{n_D} w_i \int_0^{\xi_i^s} q_z^{\xi_i} \left(\mathcal{L}_y^{\xi_i} + y_i \right) d\xi_i, \quad (45a)$$

$$M_{ya}(\mathbf{c}, t) = \sum_{i=1}^{n_D} w_i \int_0^{\xi_i^s} q_z^{\xi_i} \left(\mathcal{L}_x^{\xi_i} + \ell_i/2 - \xi_i \right) d\xi_i, \quad (45b)$$

$$M_{za}(\mathbf{c}, t) = \sum_{i=1}^{n_D} w_i \int_0^{\xi_i^s} q_y^{\xi_i} \left(\mathcal{L}_x^{\xi_i} + \ell_i/2 - \xi_i \right) - q_x^{\xi_i} \left(\mathcal{L}_y^{\xi_i} + y_i \right) d\xi_i, \quad (45c)$$

- Sliding forces

$$F_{xs}(\mathbf{c}, t) = \sum_{i=1}^{n_D} w_i \int_{\xi_i^s}^{\ell_i} \mu_d q_z^{\xi_i} d_{tx}^{\xi_i} d\xi_i, \quad (46a)$$

$$F_{ys}(\mathbf{c}, t) = \sum_{i=1}^{n_D} w_i \int_{\xi_i^s}^{\ell_i} \mu_d q_z^{\xi_i} d_{ty}^{\xi_i} d\xi_i, \quad (46b)$$

$$F_{zs}(t) = \sum_{i=1}^{n_D} w_i \int_{\xi_s^i}^{\ell_i} q_z^{\xi_i} d\xi_i, \quad (46c)$$

• Sliding moments

$$M_{xs}(\mathbf{c}, t) = \sum_{i=1}^{n_D} w_i \int_0^{\xi_s^i} q_z^{\xi_i} \left(\mathcal{L}_y^{\xi_i} + y_i \right) d\xi_i, \quad (47a)$$

$$M_{xs}(\mathbf{c}, t) = \sum_{i=1}^{n_D} w_i \int_0^{\xi_s^i} q_z^{\xi_i} \left(\mathcal{L}_x^{\xi_i} + \ell_i/2 - \xi_i \right) d\xi_i, \quad (47b)$$

$$M_{zs}(\mathbf{c}, t) = \sum_{i=1}^{n_D} w_i \int_0^{\xi_s^i} \mu_d^{\xi_i} q_z^{\xi_i} \left(d_{ty}^{\xi_i} \left(\mathcal{L}_x^{\xi_i} + \ell_i/2 - \xi_i \right) - d_{tx}^{\xi_i} \left(\mathcal{L}_y^{\xi_i} + y_i \right) \right) d\xi_i, \quad (47c)$$

where n_D is the total number of ribs. Integrals (44) and (45) can be analytically solved. Conversely, integrals (46) and (47) must be numerically calculated.

3.3. The transition coordinate

Starting from the beginning of each rib contact region, the bristle tips stick to the ground up to the transition coordinate ξ_s^i . Determining ξ_s^i entails locating the precise point where the local force induced by bristle deformation in adhesion equals the local maximum static friction force that can be sustained. This requires applying the adhesion criteria outlined in (18)

$$\left(q_x^{\xi_i} \right)^2 + \left(q_y^{\xi_i} \right)^2 = \left(\mu_s q_z^{\xi_i} \right)^2. \quad (48)$$

Due to the complexity of the pressure distribution model and the shear field description, finding an analytical solution to Eq. (48) is not feasible. However, through calculus, it can be demonstrated that the longitudinal and lateral tangential stresses $q_x^{\xi_i}$, $q_y^{\xi_i}$, and normal stress $q_z^{\xi_i}$ can be described by polynomial functions. Consequently, both the right- and left-hand sides of (48) are also polynomials. We can therefore exploit many powerful and robust root-finding algorithms, like Sturm's sequence and the Bisection methods [5] or the Algorithm 748 by Alefeld et al. [1].

3.4. Tire nonlinear system and solution method

As mentioned earlier, tire behavior is greatly influenced by carcass compliance. However, most tire models similar to the one presented in this paper (i.e., Sakai's [59–62], NEO-FIALA [26,27,42–44], TREADSIM [7], and TAMETIRE® [13]) lack or miss the sufficient discussion on the numerical method's performance in computing carcass deformation. This is a critical aspect, as the carcass deformation is the most challenging part of the tire model to tackle. Indeed, to solve the nonlinear system arising from carcass modeling (11), an efficient iterative approach satisfying real-time constraints is necessary. Many tire models, such as those in [19,20,26,27,42–44], employ Picard's Iteration Method (also known as the Fixed-Point Method), which is simple to implement but suffers from a slow convergence rate, potentially compromising model robustness. Alternatively, TAMETIRE® employs a mixed Newton/quasi-Newton iterative method [13]. If the Jacobian $\mathbf{J}_G^1$ is unavailable, *quasi-Newton* methods can be utilized, albeit with a slight decrease in convergence rate and robustness. Notice that the Frobenius norm² of the Jacobian $\mathbf{J}_F^3$ is typically small ($\|\mathbf{J}_F\|_F \ll 1$), while the Frobenius norm of the carcass stiffness matrix Jacobian $\mathbf{J}_K$ is usually large ($\|\mathbf{J}_K\|_F \gg 1$). Therefore, a good first approximation of $\mathbf{J}_G$ is given by

$$\mathbf{J}_G = \mathbf{J}_K - \mathbf{J}_F \approx \mathbf{J}_K. \quad (49)$$

If we set the initial iteration point at $\mathbf{c} = [0, 0, 0]^T$, the bottoming effect of the carcass stiffness is absent. As a consequence, the initial estimation of the Jacobian $\mathbf{J}_G$ can be straightforwardly computed by accounting solely for the structural and inflation pressure components of the carcass stiffness.

3.5. Notes on tire model implementation

The tire model presented in this paper is designed to be implemented in a straightforward manner, with the aim of being used in real-time applications. To achieve this, the model must be robust, accurate, and efficient. In this section, we discuss the robustness of the model and the strategies adopted to address potential sources of instability, as well as some software implementation details.

¹ $\mathbf{J}_G \stackrel{\text{def}}{=} \mathbf{J}_G(\mathbf{c}) = \partial \mathbf{G}(\mathbf{c}) / \partial \mathbf{c}$.

² For a matrix $\mathbf{A} \in \mathbb{R}^{n \times m}$, the Frobenius norm is defined as $\|\mathbf{A}\|_F = \sqrt{\sum_{ij} A_{ij}^2}$.

³ $\mathbf{J}_F \stackrel{\text{def}}{=} \mathbf{J}_F(\mathbf{c}) = \partial \mathbf{F}(\mathbf{c}) / \partial \mathbf{c}$.

3.5.1. Sources of instability and their solutions

Instabilities of parabolic carcass tire models are not a comprehensively treated topic, with the only two analyses being found in Pacejka [48] and in de Hoogh [7]. Although the present model is robust in ordinary and heavy-duty operating ranges, in its development the authors encountered instabilities that have not been addressed nor directly mentioned in the literature and that have been consistently resolved with the strategies described below.

1. The first issue is *big carcass deformations* caused by low stiffness or extreme vertical loads, which can result in unrealistic solutions. To address this problem, we saturated the deformation of the tire carcass by properly introducing a bottoming effect.
2. The second issue is the *transition to full-slide* regime, which can cause significant changes in the Jacobian $\mathbf{J}_G$ and possible oscillations around the solution. However, the adoption of proper nonlinear solvers can avoid this issue.

These two sources of instability are not discussed in the literature so far, but they are important to be considered in the design of a tire model for real-time applications.

3.5.2. Software implementation details

While this paper does not delve into an exhaustive discussion of the software, it is crucial to highlight that implementing the model is no simple task. It involves numerous optimizations to ensure speed, accuracy, and robustness on real-time platforms. Specifically, the model is coded in C++ (2011 standard), using the EIGEN linear algebra template library [21] and the tire-ground contact model presented in [72,73]. The code follows a modular structure, and it is divided into three main parts.

- *Tire model*: contains the model implementation and the classes that implement the various parts of the model, i.e., ribs, carcass, tread, friction coefficient model, and many other minor components.
- *Solver*: contains the implementation of the numerical methods.
- *Tire-ground contact*: contains the implementation of the tire-ground contact model as an external C++ library [73].

Fig. 8 shows the flowchart of the implemented solution algorithm.

4. Numerical performance and validation

In this section, we present numerical simulations to validate the tire model and the numerical solution approach. Specifically, we will compare the performance of the quasi-Newton methods used to solve the nonlinear system of Eqs. (11), as well as the fitting quality of the tire model to experimental data. A comparison with the MAGIC FORMULA is also provided.

4.1. Performance of the quasi-Newton methods

We evaluate the performance of the quasi-Newton numerical methods mentioned earlier. Specifically, we aim to identify the most suitable method for solving the nonlinear system of Eqs. (11) among the following:

- Picard Iteration Method (PIM);
- Greenstadt's 1st Method (G1M) [71];
- Greenstadt's 2nd Method (G2M) [71];
- Eirola-Nevanlinna Method (ENM) [10];
- Broyden's Bad Method (BBM) [4];
- Broyden's Good Method (BGM) [4];
- Broyden's Combined Method (BCM) [38].

and their respective dumped versions, denoted with the prefix “D-”. We will compare these algorithms based on function evaluations, iterations, convergence speed, and success ratio.

Table 1 presents the performance data of the quasi-Newton solvers. The tests are conducted on a set of 10^5 simulations with inputs distributed evenly within the following ranges $\sigma_x \in [-1, 1]$, $\sigma_y \in [-1, 1]$, $\varphi \in [-0.1, 0.1]$, $F_z \in [0, 5 \cdot 10^5]$, and $\gamma \in [-\pi/10, \pi/10]$, which represent typical operating conditions for a passenger car tire. All tests are performed on a shielded CPU. The simulation platform is a Concurrent® iHawk™ with 2.5 GHz Intel® Xeon® Silver 4215 8-core, 11 MB cache, 32 GB DDR4 RAM, and 64 bit RedHawk™ Linux real-time operative system with CentOS distribution.

As depicted in Table 1, both the PIM and D-PIM fail to converge to the solution with the required tolerance. The G1M and D-G1M cannot complete the tests due to overflow issues. Conversely, the G2M and ENM, and their dumped version, D-G2M and D-ENM, show a good success ratio but exhibited high average and variance iterations and function evaluations. Although the pure BBM suffered from overflow issues, its dumped version, D-BBM, successfully converged to the solution with the required tolerance and maintained an acceptable success ratio. The BGM and BCM, as well as their dumped versions, D-BGM and D-BCM, demonstrated the best performance overall. They exhibited the highest success ratio and the lowest average and variance in both iterations and function evaluations. Notably, the BCM showed superior computational efficiency. It is worth mentioning that no relaxation steps were accepted in D-BCM. Consequently, the performance of the BCM method is deemed excellent for meeting the demands of hard real-time simulations, i.e., achieving a computational time of less than 1 ms. Only a small percentage of the tests required and a high

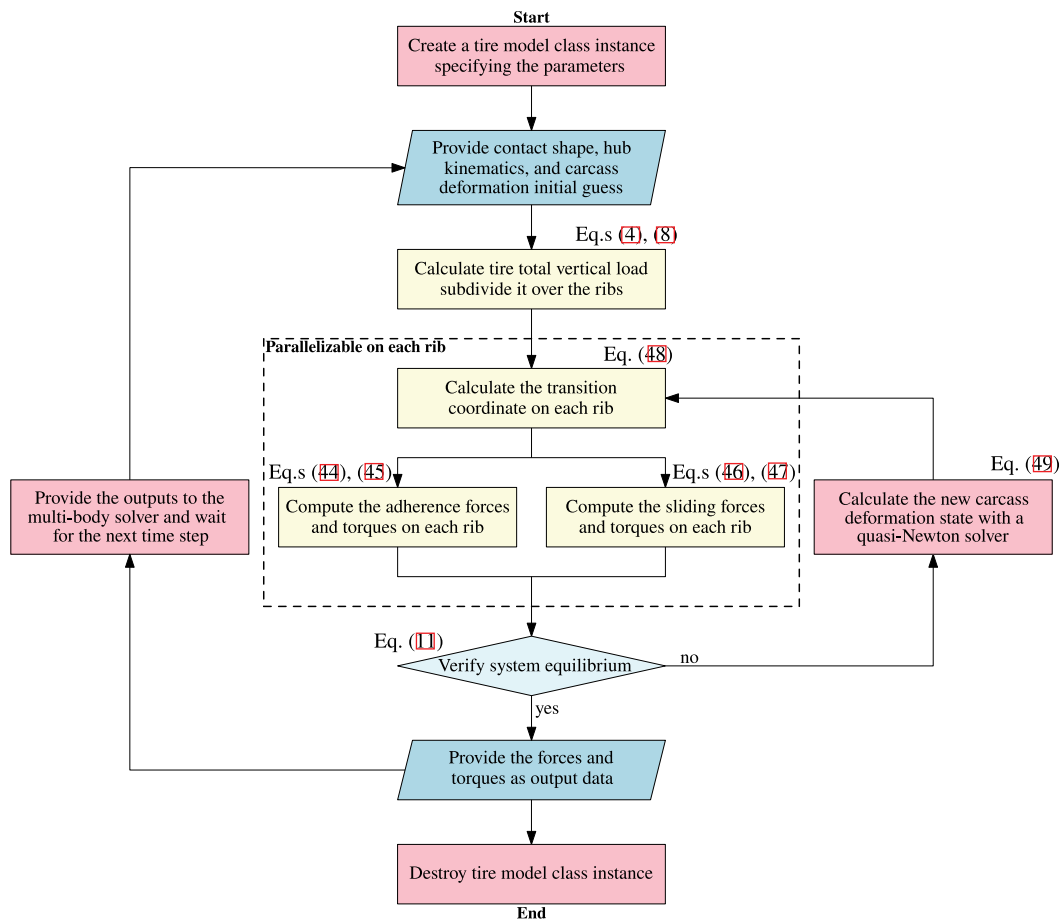


Fig. 8. Flowchart of the implemented solution algorithm.

number of iterations, however, in these cases the vertical load and resulting carcass deformations exceeded the structural limits, leading to instability in the iterative process.

The bottom part of Table 1 presents the performance of the BCM in relation to the number of ribs. Notably, the number of ribs does not impact the efficacy of the iterative method but merely affects the computational time. This arises from the fact that a change in the number of ribs does not alter the number of unknowns in (11). Consequently, the computational time scales linearly with the number of ribs. However, it is worth noting that the variance of the computational time increases more than linearly due to small yet significant variations in the performance of Algorithm 748.

4.2. Validation of the tire model

The validation of the proposed tire model is performed by minimizing the Root Mean Square (RMS) error between its predictions and experimental data. Similar to the MAGIC FORMULA, the proposed tire model undergoes sequential fitting. Initially, the vertical characteristic is fitted, followed by the linear behavior of the tire model under small pure-slip conditions. Within this linear region and across a wide range of vertical loads, the tread shear stiffness and carcass deformation stiffness are adjusted. This fitting procedure is supported by studies in [26,27,42–44], illustrating how carcass deformation mechanics impact the tire's linear behavior during cornering. Similar results are reported in Fig. 9, where we depict the influence of vertical load on lateral force and self-aligning moment generation, highlighting carcass deformation's effect on the cornering behavior. The fitting of the tire model under large- and combined-slip conditions is performed last, adjusting the tire model's nonlinear behavior through the friction coefficient model.

To demonstrate the effectiveness of the proposed fitting procedure, we validate the proposed tire model under pure slip conditions by comparing its predictions with those of a MAGIC FORMULA 5.2 [49] on the fitting of experimental data for a HOOSIER® 18 × 6-10 LCO tire. As previously mentioned in the introduction, the dataset is obtained from the Calspan Tire Research Facility (TIRF) for the Formula SAE Tire Test Consortium (FSAE TTC) testing program [34]. In Fig. 11, we compare the fitting of the longitudinal and lateral forces and self-aligning moment for the HOOSIER® tire at several discrete values of vertical load with zero camber angle. The fitting parameters of the proposed tire model are reported in Table 2. The results indicate that the proposed model provides

Table 1

Comparison between quasi-Newton methods' performance based on a dataset comprising 10^5 tests. The inputs are evenly distributed within the ranges $\sigma_x \in [-1, 1]$, $\sigma_y \in [-1, 1]$, $\varphi \in [-0.1, 0.1]$, $F_z \in [0, 5 \cdot 10^5]$, and $\gamma \in [-\pi/10, \pi/10]$, using a 325/660R13 race car tire discretized with 5 ribs. Convergence is considered achieved when $\|G(c)\|_2 \leq 10^{-8}$ and $\|c_{k+1} - c_k\|_2 \leq 10^{-16}$. Maximum algorithm iteration and relaxation steps are set to 100 and 10, respectively. The bottom part of the table demonstrates the impact of changing the number of ribs on the BCM method's performance.

Quasi-Newton Methods Performance with 5 Ribs

Method	Evaluations (–)				Iterations (–)				Time (μ s)				Success
	min	max	μ	σ^2	min	max	μ	σ^2	min	max	μ	σ^2	
PIM	100	100	100.0	0.00	100	100	100.0	0.0	518.0	7949.0	617.2	13 953.2	0.0%
G1M	–	–	–	–	–	–	–	–	–	–	–	–	Overflow
G2M	4	100	9.2	55.8	4	100	9.2	55.8	23.5	1089.4	69.0	5252.0	99.5%
ENM	6	200	11.9	234.0	3	100	6.5	58.5	25.0	2025.0	81.5	11 002.8	99.4%
BBM	–	–	–	–	–	–	–	–	–	–	–	–	0.0%
BGM	4	100	8.8	18.4	4	100	8.8	18.4	22.6	948.1	62.5	1973.0	99.9%
BCM	4	17	8.4	6.7	4	17	8.4	6.7	22.9	264.4	55.7	885.9	100.0%
D-PIM	100	496	190.7	13 628.1	100	100	100.0	0.0	507.0	5383.0	1151.9	568 491	0.0%
D-G1M	–	–	–	–	–	–	–	–	–	–	–	–	Overflow
D-G2M	4	376	11.3	572.4	4	100	9.3	57.1	21.5	3195.5	80.4	40 321.9	99.5%
D-ENM	6	513	12.8	829.7	3	100	6.3	44.3	27.0	5207.7	97.5	52 752.6	99.5%
D-BBM	4	125	11.0	58.0	4	42	9.3	12.8	22.3	1284.0	76.9	5126.3	100.0%
D-BGM	4	376	9.7	110.6	4	100	8.7	15.1	21.3	3238.9	67.0	8142.9	99.9%
D-BCM	4	35	9.1	14.6	4	17	8.4	6.7	22.5	316.8	62.1	1615.1	100.0%

Broyden Combined Method Performance

Number of Ribs	Evaluations (–)				Iterations (–)				Time (μ s)				Success
	min	max	μ	σ^2	min	max	μ	σ^2	min	max	μ	σ^2	
1	6	11	7.9	10.3	6	11	7.9	10.3	12.1	89.0	19.4	20.6	100.0%
5	4	17	7.9	10.3	4	17	7.9	10.3	22.9	264.4	55.7	885.9	100.0%
10	6	11	7.9	10.3	6	11	7.9	10.3	106.9	361.8	167.3	1746.22	100.0%
15	6	11	7.9	10.3	6	11	7.9	10.3	158.0	523.5	249.4	3878.2	100.0%
20	6	11	7.9	10.3	6	11	7.9	10.3	210.1	693.3	332.9	7028.7	100.0%
25	6	11	7.9	10.3	6	11	7.9	10.3	262.7	858.4	414.0	10 708.3	100.0%

Methods legend: Fixed-Point Method (PIM), Greenstadt 1st Method (G1M), Greenstadt 2nd Method (G2M), Eirola-Nevanlinna Method (ENM), Broyden Bad Method (BBM), Broyden Good Method (BGM), and Broyden Combined Method (BCM). Dumped XX Method (D-XXM).

Symbols legend: min minimum number of evaluations/iterations, max maximum number of evaluations/iterations, μ average evaluations/iterations, and σ^2 evaluations/iterations variance.

Simulation Platform: Concurrent[®] iHawk[™] with 2.5 GHz Intel[®] Xeon[®] Silver 4215 8-core, 11 MB cache, 32 GB DDR4 RAM, and 64 bit RedHawk[™] Linux real-time operative system with CentOS distribution.

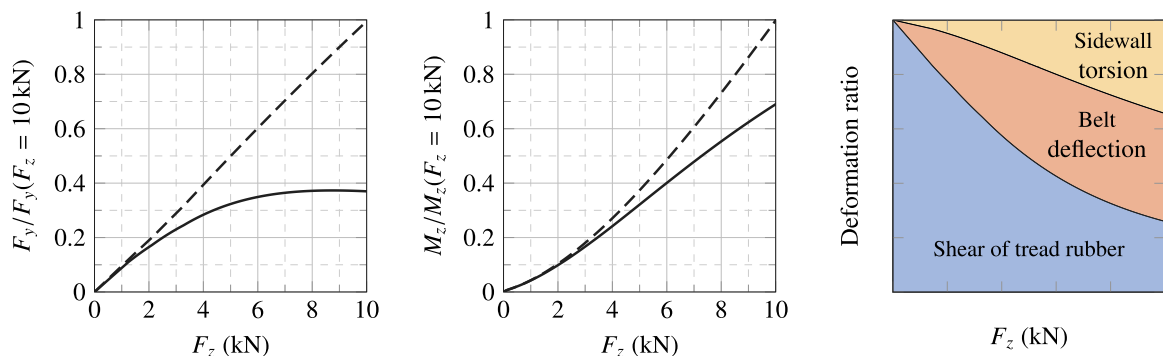


Fig. 9. Comparison between the flexible carcass and rigid carcass models of the 325/660R13 race car tire under a small pure lateral slip condition ($\alpha = \pi/180$). The results highlight the impact of carcass deformation on the tire's cornering behavior. Additionally, the *qualitative* deformation ratios of tread shear, belt deflection, and sidewall torsion are illustrated in the right-hand side plot. **Legend:** — flexible carcass tire model, and - - rigid carcass tire model.

data fitting comparable to the MAGIC FORMULA 5.2 for the lateral force, while a higher RMS error is observed for the longitudinal force and self-aligning moment. Specifically, the high RMS observed for the self-aligning moment is due to the lack of a dataset that

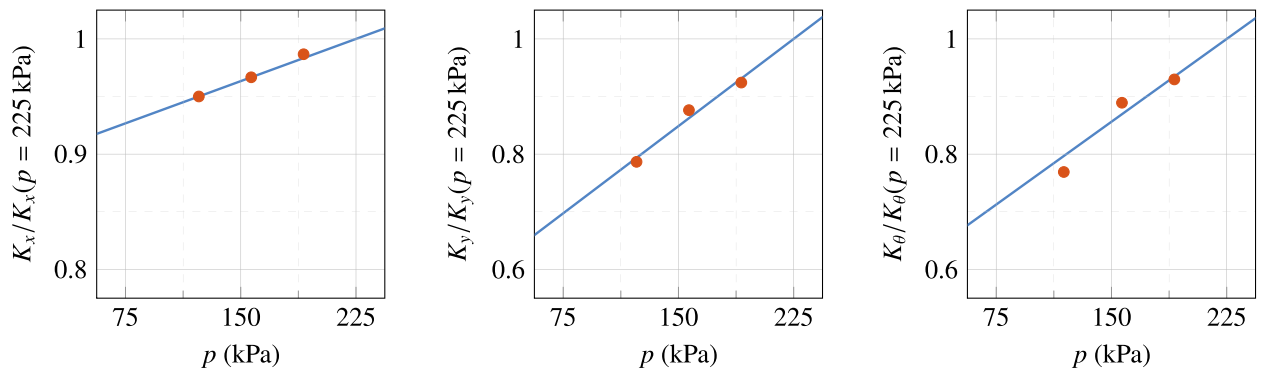


Fig. 10. Longitudinal, lateral and torsional carcass structural dependency on tire inflation pressure. The experimental data is obtained from the tire manufacturer through static deflection measurements of a 325/660R13 race car tire at three distinct inflation pressures. Each stiffness is normalized by its value of at 225 kPa to preserve confidentiality. Legend: ■ proposed carcass model, and ■ experimental data.

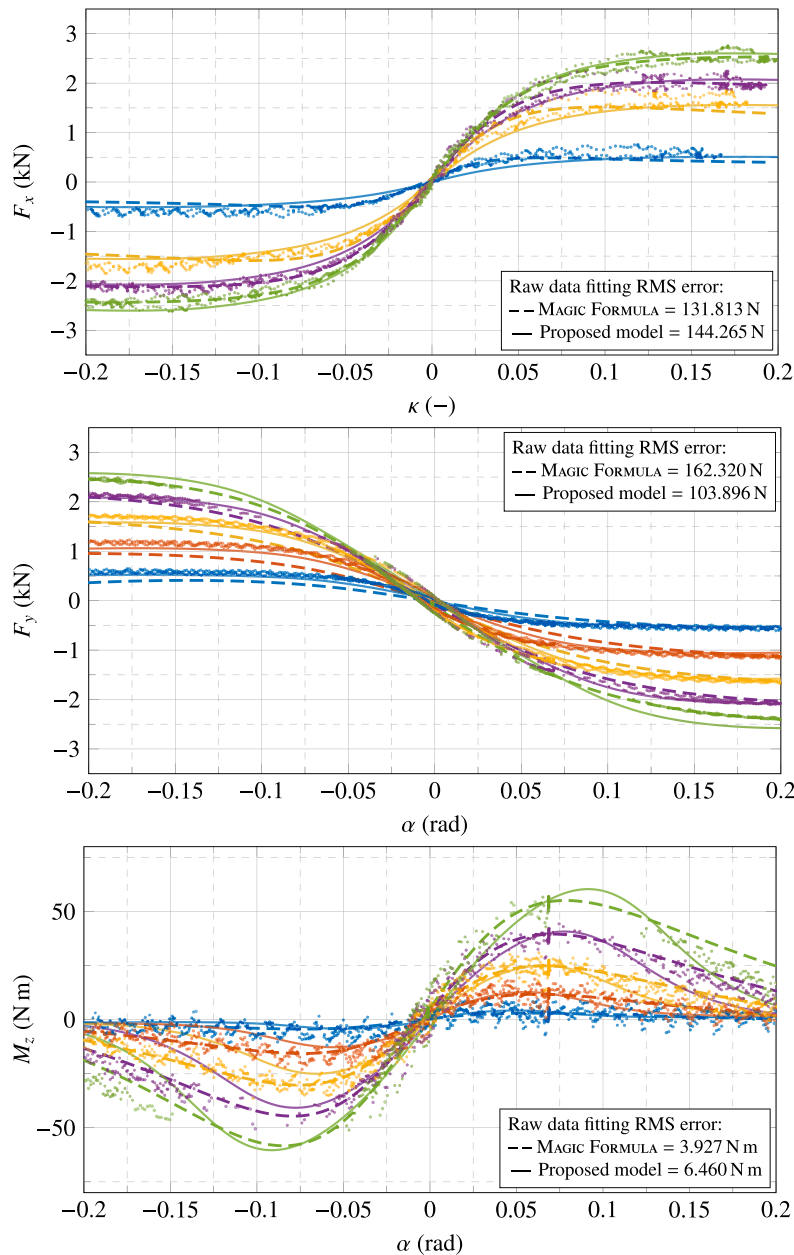
Table 2

Parameters of the proposed tire model for the HOOSIER® 18 × 6-10 LCO tire used in the fitting of the experimental data, as well as in the comparison with the MAGIC FORMULA 5.2. Parameters marked with a * symbol have not been fitted due to the lack of experimental data. Instead, they have been estimated from data available in the literature.

Symbol(s)	Value(s)	Unit	Description
k_z^s	1.451e+5*	N/m	Structural vertical stiffness
k_z^p	0.688e+0*	N/(m Pa)	Vertical stiffness dependency on inflation pressure
k_z^w	1.097e+0*	N s/(m rad)	Vertical stiffness rolling speed dependency
k_z^b	3.585e+4*	N/(m rad)	Vertical stiffness camber angle dependency
k_z^b	1.000e+8*	N/m	Vertical bottoming stiffness
ρ_b	7.600e-2*	m	Tire bottoming deflection
c_z	2.000e+2*	N s/m	Vertical damping
k_x^s	6.500e+3*	N/m	Carcass longitudinal structural stiffnesses
k_x^p	9.000e-1*	N/(m Pa)	Carcass longitudinal stiffnesses pressure dependency
k_x^b	1.000e+7*	N/m	Carcass longitudinal bottoming stiffnesses
x_b	5.000e-2*	N/m	Carcass longitudinal bottoming deflection
k_y^s	1.000e+4*	N/m	Carcass lateral structural stiffnesses
k_y^p	1.210e-1*	N/(m Pa)	Carcass lateral stiffnesses pressure dependency
k_y^b	1.000e+7*	N/m	Carcass lateral bottoming stiffnesses
y_b	5.500e-2*	N/m	Carcass lateral bottoming deflection
k_θ^s	1.000e+3*	N/rad	Carcass torsional structural stiffness
k_θ^p	5.750e-2*	N/(rad Pa)	Carcass torsional stiffness pressure dependency
k_θ^b	1.000e+6*	N/rad	Carcass torsional bottoming stiffness
θ_b	1.200e-1*	N/rad	Carcass torsional bottoming deflection
Ψ	2.943e+0*	–	Carcass bending shape factor
λ	4.046e+0*	–	Contact pressure distribution convexity parameter
δ	1.993e-3*	–	Contact pressure distribution barycentre shift
ν	1.000e+0*	–	Tread void ratio
k_{cx}, k_{yy}	3.511e+7, 5.508e+7	N/m ³	Bristles longitudinal stiffnesses
k_{xy}, k_{yx}	0.000e+7*	N/m ³	Bristles cross stiffness components
μ_w	1.000e+0*	–	Friction coefficient scaling factor
μ_s, μ_k	2.634e+0, 2.598e+0	–	Static and kinetic friction coefficients
v_μ	2.060e+0*	m/s	Friction Stribeck velocity
$\mu_{p_0}, \mu_{p_1}, \mu_{p_2}$	0.201e+0, 7.109e-1, 7.043e-7	–	Friction coefficient vertical pressure dependency
μ_{v_0}, μ_{v_1}	1.034e+0, 0.101e+0	–	Friction coefficient sliding speed dependency

comprises a wider range of working conditions. This leads to wrong parameter estimation, as the full range of frictional behavior is not captured, and as a result, the model is not able to accurately predict the overall behavior of the tire. Nonetheless, it is also important to note that the longitudinal, lateral, and self-aligning moment forces are strongly coupled, and the fitting of one force can affect the fitting of the others. This is particularly true for the self-aligning moment, which is highly dependent on both frictional and structural properties (refer also to [45,65]). Given the limited number of parameters and the simplicity of the proposed model, these results are considered satisfactory.

To further validate the proposed tire model, we also assess its capability to reproduce the behavior of a 325/660R13 race car tire under pure and combined slip conditions. The data is provided by the tire manufacturer via a MAGIC FORMULA 5.2 tire property file (*.tir). Fig. 10 illustrates the fitting of the longitudinal, lateral and torsional carcass stiffnesses on inflation pressure within the linear region of the carcass deformation. Stiffness data is obtained from the tire manufacturer through static deflection measurements at three distinct inflation pressures. Each stiffness is normalized by its value at 225 kPa to preserve confidentiality. Figs. 12 and 13



Pure Slip Properties

Tire: HOOSIER® 18x6-10 LCO



Road: Flat

Experimental setup at the Calspan Tire Research Facility (TIRF) for the Formula SAE Tire Test Consortium (FSAE TTC) testing program [50]. Picture from the documentation accompanying the FSAE TTC “Round 5” dataset.



Fig. 11. Comparison between the MAGIC FORMULA and the proposed model regarding the fitting of experimental data for a HOOSIER® 18 × 6-10 LCO tire. Experiments are performed under pure longitudinal and lateral slips at several discrete values of vertical load with zero camber angle. The dataset is provided by the Calspan Tire Research Facility (TIRF) for the Formula SAE Tire Test Consortium (FSAE TTC) testing program [34]. *Lines and marks legend:* • experimental data, — proposed model, and --- MAGIC FORMULA 5.2. *Colors legend:* ■ $F_z = 222$ N, ■ $F_z = 444$ N, ■ $F_z = 667$ N, ■ $F_z = 890$ N, and ■ $F_z = 1120$ N.

report the results for pure and combined slips conditions, respectively. Camber thrust effects are assessed in Fig. 14 for the same 325/660R13 tire rolling over a cobbled road. In this case, zero longitudinal and lateral slips are enforced and the camber angle varies due to road macro-asperities. On these last three plots, y-axis labels are hidden to preserve data confidentiality. As depicted, the proposed model provides a data fitting that is comparable to that provided through the MAGIC FORMULA 5.2 for all components in both pure and combined slip conditions, as well as for the camber thrust effect. In this case, the dataset is more comprehensive, and it ranges from low to high vertical loads, thus allowing for clearer parameter estimation. However, it is evident from Fig. 13 that there is a slight discrepancy in the fitting of the self-aligning moment under small combined slip conditions. Similarly, Fig. 14 shows that the self-aligning moment arising from uneven road surfaces is not accurately captured at high vertical loads. This is due

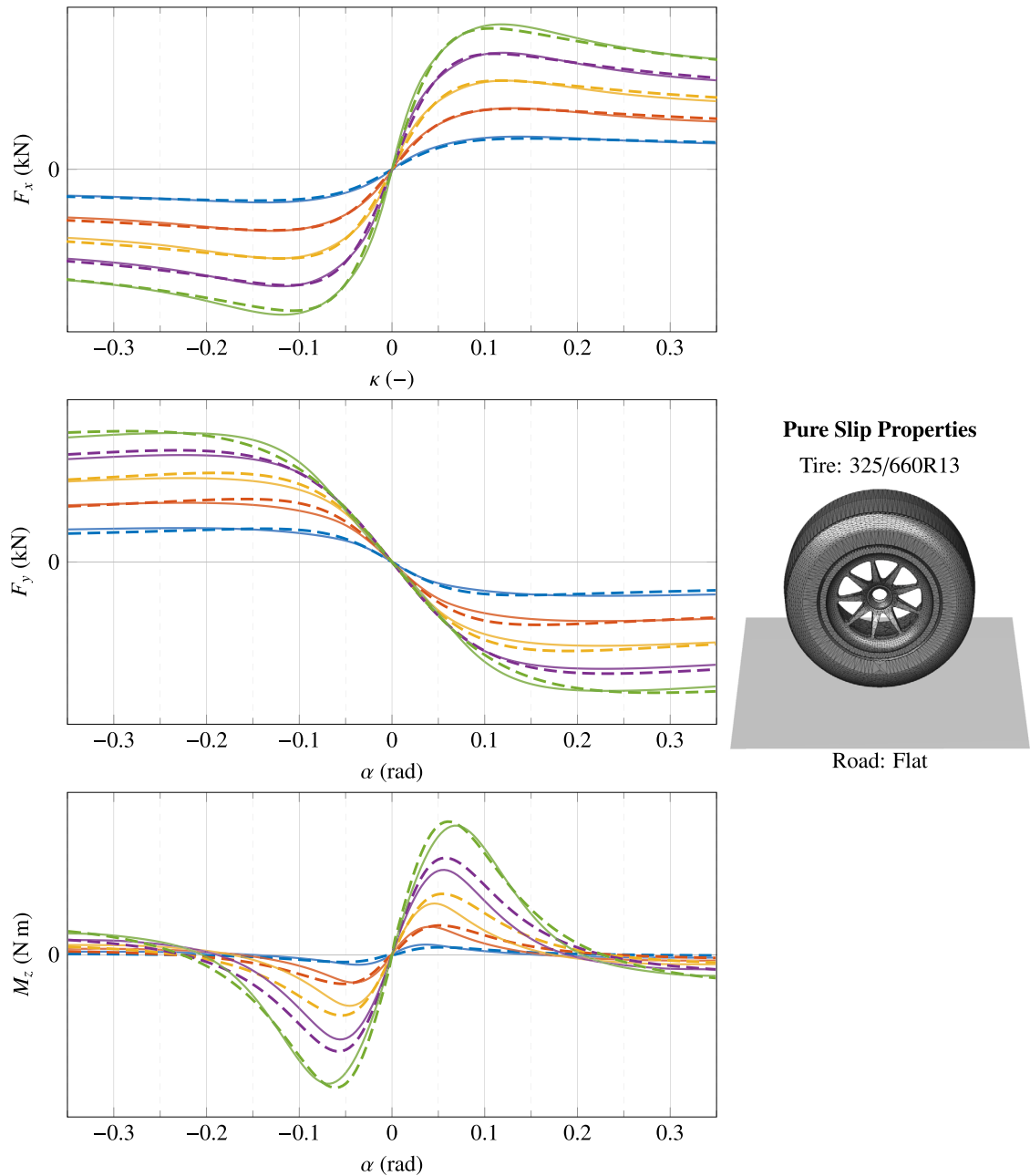


Fig. 12. Comparison between the MAGIC FORMULA and the proposed model for a 325/660R13 race car tire under pure slips conditions. Results are reported for different vertical loads with zero camber angle. Ordinate axes labels are removed to respect data confidentiality. Lines and marks legend: — proposed model, and - - MAGIC FORMULA 5.2. Colors legend: ■ $F_z = 2$ kN, ■ $F_z = 3$ kN, ■ $F_z = 4$ kN, ■ $F_z = 5$ kN, and ■ $F_z = 6$ kN.

to the carcass nonlinear effects, which subtly affect the tire's behavior in these conditions, as well as nontrivial pressure distribution emerging from the road irregularities [45].

5. Conclusions and future works

In this paper, we have introduced a novel tire brush model characterized by flexibility and optimized for real-time applications while maintaining the physical significance of its parameters. Carcass deformations, including in-plane fore-aft displacement, in-plane deflection, and out-of-plane sidewall torsion, are approximated through a second-order polynomial. Forces and torques arising

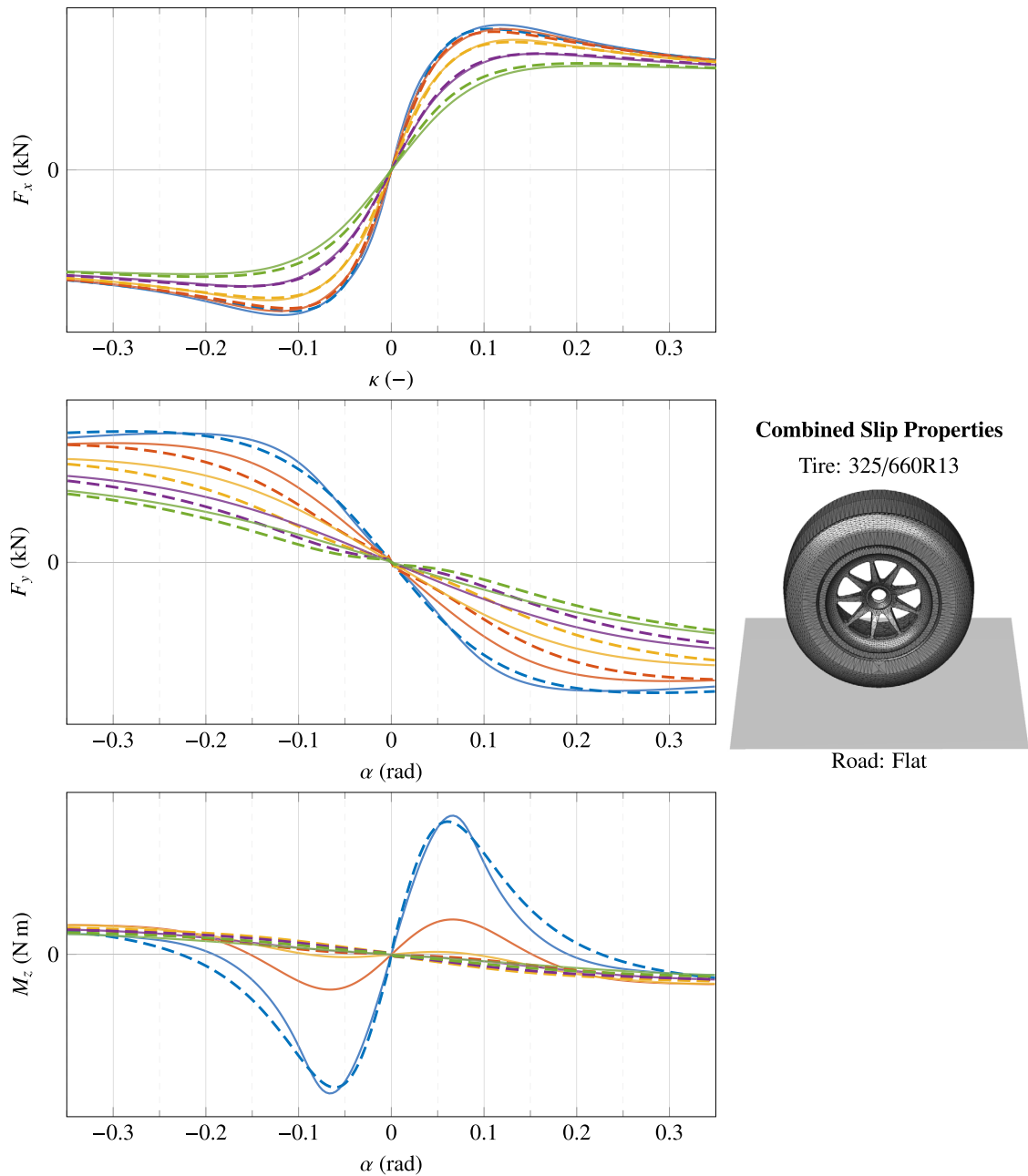


Fig. 13. Comparison between the MAGIC FORMULA and the proposed model for a 325/660R13 race car tire under combined slips conditions. Results are reported for a vertical load of $F_z = 5$ kN and zero camber angle. Ordinate axes labels are removed to respect data confidentiality. *Lines and marks legend:* — proposed model, and - - MAGIC FORMULA 5.2. *Colors legend:* ■ $\kappa = 0$, ■ $\kappa = 0.1$, ■ $\kappa = 0.2$, ■ $\kappa = 0.3$, and ■ $\kappa = 0.4$ for longitudinal slip. ■ $\alpha = 0$ rad, ■ $\alpha = 0.035$ rad, ■ $\alpha = 0.70$ rad, ■ $\alpha = 0.105$ rad, and ■ $\alpha = 0.140$ rad for lateral slip.

from tire–road contact are described using brush mechanics, with bristle kinematics directly linked to carcass deformation. Saturation effects are introduced to prevent unrealistic carcass deformations, and the tire’s nonlinear behavior is captured through Savkoor’s friction model with both pressure and velocity dependencies. To tackle the complexity arising from the formulation of the tire model, we have discretized the tire’s geometry into ribs, each independently intersecting the local road surface, enabling both accurate representation and computational efficiency. The model’s nonlinear system of equations arising from the carcass deformation and tire–road contact equilibrium has been derived and discussed, highlighting the challenges of solving it. Specifically, considerations on the Jacobian estimation and quasi-Newton method initialization have been presented, demonstrating the robustness and efficiency of

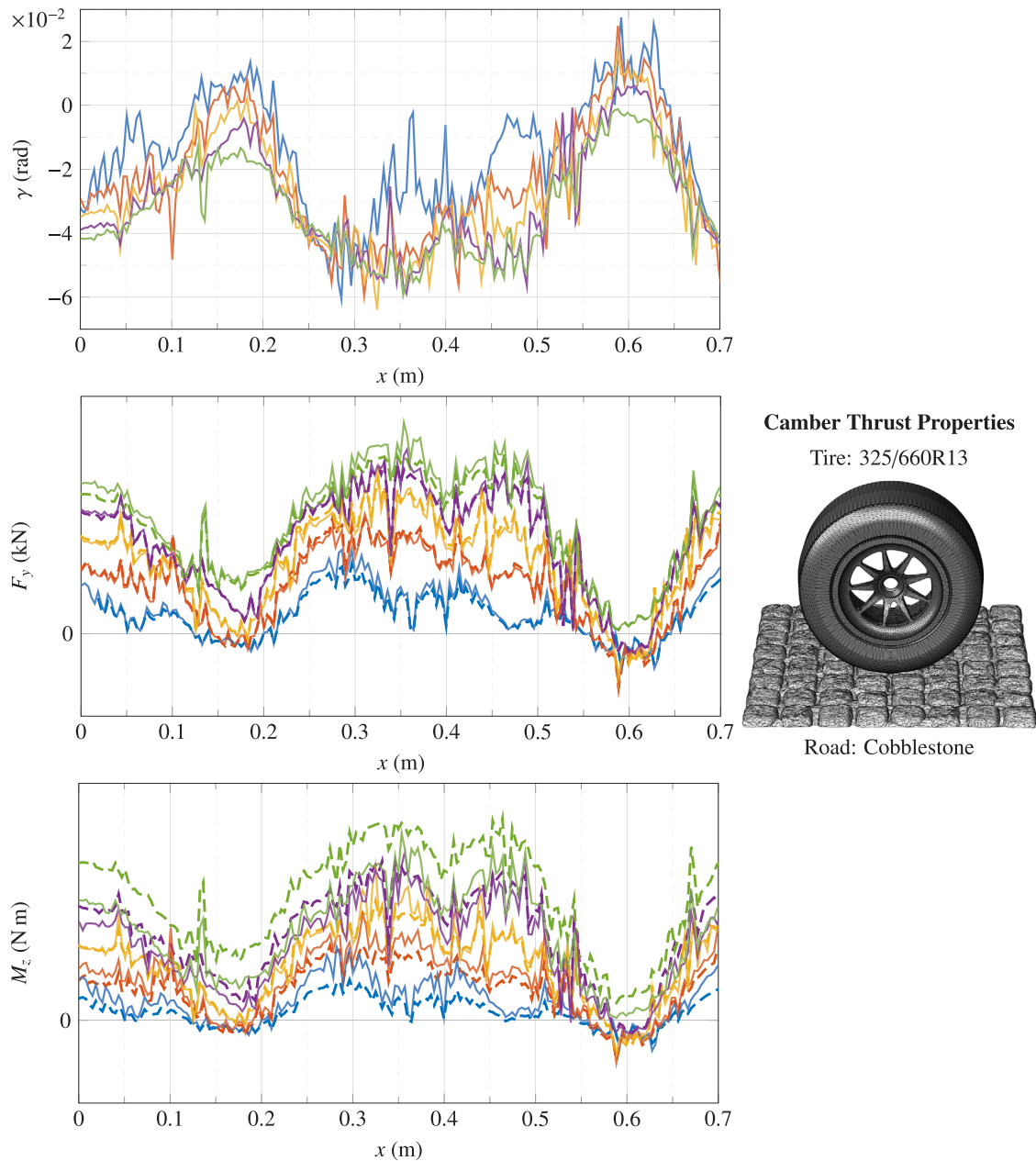


Fig. 14. Comparison between the MAGIC FORMULA and the proposed model for a 325/660R13 race car tire rolling over a cobblestone road. Results are reported for zero longitudinal and later slips, at different vertical loads. Ordinate axes labels are removed to respect data confidentiality. *Lines and marks legend:* — proposed model, and - - MAGIC FORMULA 5.2. *Colors legend:* ■ $F_z = 2$ kN, ■ $F_z = 3$ kN, ■ $F_z = 4$ kN, ■ $F_z = 5$ kN, and ■ $F_z = 6$ kN.

the proposed solution algorithm. Sources of instability have been identified and discussed, providing a comprehensive understanding of the model's behavior, as well as the strategies employed to address them. Finally, we have developed an efficient numerical solver based on quasi-Newton solvers.

The model has been tested and validated under a wide range of operating conditions, including low inflation pressure, low rolling speed, and low/high vertical loads. Nonetheless, the quasi-Newton Broyden's Combined Method has proven to ensure the real-time performance of the model. A computational time of less than 1 ms is achieved in a real-time simulation platform, with a full success ratio, thereby demonstrating the model's suitability for real-time applications. Scalability with the number of ribs has also been demonstrated, enabling the model to be used in a wide range of applications and with different levels of computational resources. Carcass stiffness and inflation pressure dependency have been successfully fitted to experimental data, demonstrating the efficacy of

the carcass model despite its simplicity. A sequential parameter fitting procedure based on tire mechanics and considerations from the literature and physical insights, for the minimization of the RMS error between the proposed tire model and experimental data, has been presented. The results demonstrate its accuracy in predicting the tire's behavior under pure and combined slip conditions, as well as in handling camber thrust effects. However, the fitting of the self-aligning moment is not as accurate as the fitting of the longitudinal and lateral forces. This is attributed to the lack of a dataset comprising a wider range of working conditions for the HOOSIER[®] 18 × 6-10 LCO tire, as well as strong nonlinear effects in the case of the 325/660R13 tire, thus leading to small discrepancies in the behavior prediction. Nonetheless, the proposed model provides a satisfactory fitting of the experimental data, considering its very limited number of parameters, as well as its real-time performance and physical significance.

In future works, we aim to extend the model by integrating thermal and wear mechanisms, thereby enhancing the overall assessment of vehicle performance across diverse operating conditions. Moreover, our model presents an opportunity to delve deeper into the influence of design and operational parameters on tire behavior. By systematically varying parameters, we can gain a deeper understanding of their individual and collective effects on tire performance. Therefore, optimizing tire design and operational strategies to meet specific performance criteria, whether in terms of handling, traction, or fuel efficiency.

CRedit authorship contribution statement

Davide Stocco: Conceptualization, Investigation, Methodology, Software, Validation, Writing – original draft, Writing – review & editing, Formal analysis. **Francesco Biral:** Conceptualization, Formal analysis, Investigation, Methodology, Supervision, Validation, Writing – review & editing, Project administration. **Enrico Bertolazzi:** Formal analysis, Methodology, Project administration, Software, Supervision, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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